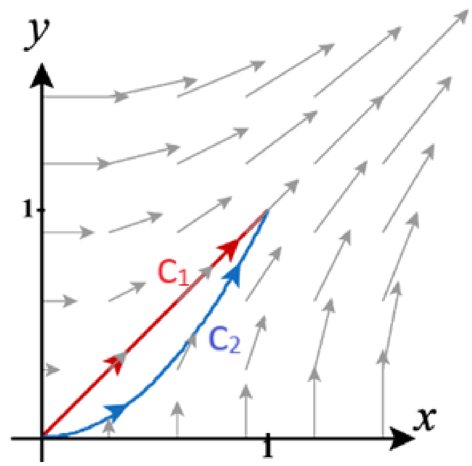
5.3 Independence of Path; Conservative Vector Fields

We begin this section with an example which will set the stage for what we will be discussing.

Example 1: Suppose C_1 is given by $\mathbf{r}_1(t)=\langle t,t\rangle$ where $0\le t\le 1$. Suppose C_2 is given by $\mathbf{r}_2(t)=\langle t,t^2\rangle$ where $0\le t\le 1$.

a) Where do the above curves start and end?

b) Show $\int_{C_1} ydx+xdy=\int_{C_2} ydx+xdy$.



Question: If we were to use any smooth curve, $\mathbf{r}(t)$ *from (0,0) to (1,1), would we still get that* $\int_C \langle y,x\rangle\cdot d\mathbf{r}=1$*?*

Questions like the one posed above are the theme of this section. Surprisingly it is the case that for some vector fields the path that we take from point A to point B has *no effect* on the value of the resulting line integral! This is a FASCINATING idea especially considering the fact that there are infinite number of paths from A to B and the complexity and length of some of these paths can be astounding!

So, if we have the right kind of vector field computing a line integral will ultimately only depend on the starting point and ending point for our object. What do we mean though by the "right kind of vector field". It turns out that the in order for the path of motion to not have any affect on the resulting line integral we simply must be working with a *conservative* vector field. The proof of the next theorem shows us why this is the case.

Theorem (Fundamental Theorem of Line Integrals):
Let *C* be a smooth curve given by a vector function $\mathbf{r}(t)$ where $a \le t \le b$. Let f be a differentiable function of two or three variables whose gradient vector $\vec{\nabla} f$ is continuous on *C*. Then,

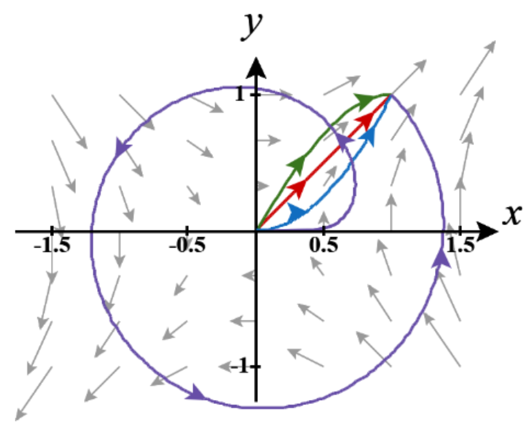
$$\int_C \vec{\nabla} f \cdot d\mathbf{r} = f\big(\mathbf{r}(b)\big) - f\big(\mathbf{r}(a)\big).$$

Note: This result is amazing! It says that if our vector field of interest is conservative (i.e. is the gradient of some scalar function) all we need to know about the curve to compute the line integral is the endpoints!

Proof (may be omitted from class): We will prove this in the case that **r** has three components. We know that by definition: $\int_C \vec{\nabla} f \cdot d\mathbf{r} = \int_a^b \vec{\nabla} f\big(\mathbf{r}(t)\big)\cdot \mathbf{r}'(t)\,dt = \int_a^b \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt = \int_a^b \left(\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \right) dt.$

Fact: One can generalize the above argument and show that the above result also holds if *C* is a piecewise smooth curve.

<u>Example 2:</u> Find the work done by $\mathbf{F}(x,y) = \langle y, x \rangle$ on a particle which moves along some piecewise smooth curve *C* that starts at $(0,0)$ and ends at $(1,1)$.



Note that the paths in the picture above could all possibly represent the curve C and are also not the only paths that could be the same as C.

Note: When applying the fundamental theorem of line integrals the most challenging part is finding a scalar function whose gradient equals the vector field of interest (in some cases such a scalar function might not even exist). This challenge will be discussed in much more detail later in this section.

Making Connections

Recall that a line integral was interested in "adding up" a quantity as we moved along a curve. It is easy to see that this concept extends upon the idea of definite integrals from Calculus I.

Definite Integral vs Line Integrals:

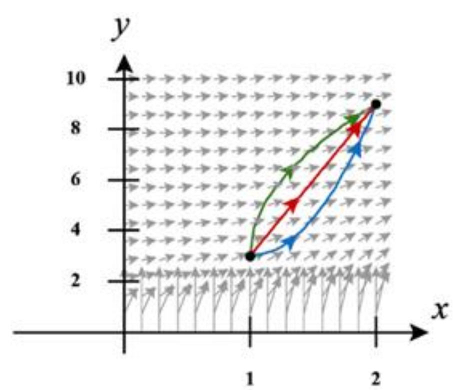
In Calculus I we had a function $f(x)$ and we were interested in "adding up" the values of this function over an interval $[a,b]$. So the integral $\int_a^b f(x)\,dx$ is very similar to the line integral $\int_C f(x,y)\,ds$.

Fundamental Theorem of Calculus vs FTOLI:

Remember that in Calculus I the Fundamental theorem showed us that to find $\int_a^b f(x)\,dx$ we can simply look at the behavior of an antiderivative at the end points of the interval $[a,b]$. Note that the FTOLI states that we can evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ by evaluating the "anti-gradient" of **F** at the endpoints of *C*!

It is AMAZING that this concept of finding an integral by evaluating a specific function at some endpoints extends across definite integrals and line integrals, however we will see even later in Chapter 5 how this concept continues to extend to specific double and triple integrals!

Example 3: Suppose *C* is a smooth curve which starts at $(1,3)$ and ends at $(2,9)$. Suppose $\mathbf{F}(x,y)=\left\langle y^2e^{xy}, e^{xy}(1+xy)\right\rangle$. Find the exact value of $\int_C \mathbf{F}\cdot d\mathbf{r}$.



We now discuss the notion of independence of path which will be of great importance for the rest of the section.

Definition: We say that C is a path if it is a piecewise-smooth curve.

Definition: Suppose that $\mathbf{F}$ is a continuous vector field with domain D. We say that the line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of path if $\int_{C_1} \mathbf{F}\cdot d\mathbf{r} = \int_{C_2} \mathbf{F}\cdot d\mathbf{r}$ for any two paths C_1 and C_2 in D that have the same initial and terminal points.

Example 4: Suppose that $f(x,y,z)$ is a scalar function such that $\mathbf{F} = \vec{\nabla} f$ is continuous on the domain D. TRUE or FALSE: $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of path.

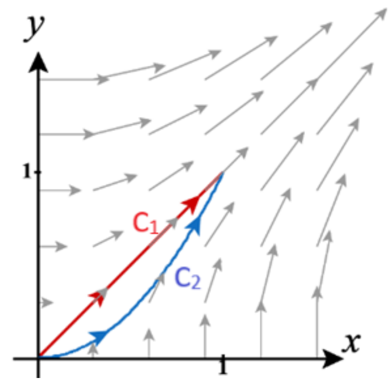
Note: Another way to say the above result is that ***line integrals of conservative vector fields are independent of path.*** *A really good question to ask from here is: are all vector fields which are independent of path conservative?*

So we have established the following fact.

Theorem (5.3.1): Suppose $\mathbf{F}$ is a continuous vector field with domain D. If $\mathbf{F}$ is a conservative vector field, then $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of path in D.

Note: The contrapositive of this statement is logically equivalent to the statement itself and hence is also true: Suppose ***F*** *is a continuous vector field with domain D. If* $\int_C \mathbf{F}\cdot d\mathbf{r}$ *is not independent of path in D, then* ***F*** *is not a conservative vector field.*

Example 5: Suppose C_1 is given by $\mathbf{r}_1(t)=\langle t,t\rangle$ where $0\le t\le 1$. Suppose C_2 is given by $\mathbf{r}_2(t)=\langle t,t^2\rangle$ where $0\le t\le 1$. Suppose that $\mathbf{F}(x,y)=\langle y^2,x^2\rangle$.



a) Find $\int_{C_1}\mathbf{F}\cdot d\mathbf{r}$.

b) Find $\int_{C_2}\mathbf{F}\cdot d\mathbf{r}$.

c) Is $\int_{C}\mathbf{F}\cdot d\mathbf{r}$ independent of path in 2 space? Explain.

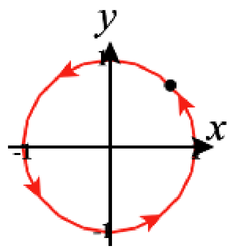
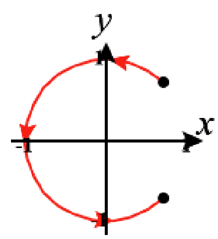
d) Is the above vector field conservative? Explain.

*Note: The idea in the above example is crucial. Specifically, it shows how we can prove a vector field is not conservative. This means in the above example it is impossible to find a scalar function whose gradient is equal to **F**.*

We have now seen that *conservative vector fields* have some very useful properties and can be used to make the calculation of line integrals much easier.

So, being able to determine if a vector field is conservative is now of great importance to us and we will spend the rest of this section trying to pursue this.

Definition: A curve is called <u>closed</u> if its terminal point coincides with its initial point.

Closed	**Non-closed**
	

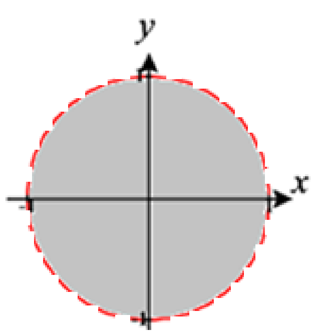
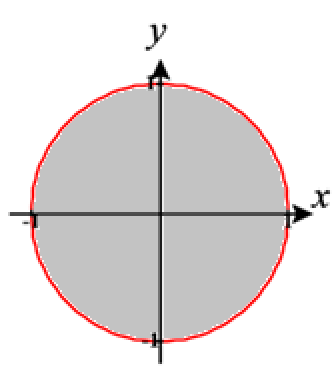
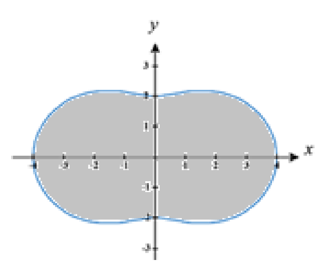
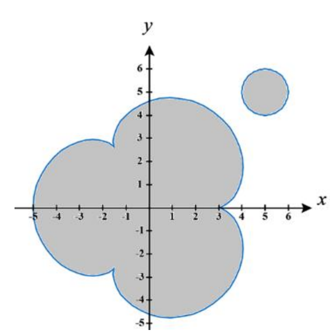
Theorem (5.3.2): Suppose that **F** is a continuous vector field with domain D. $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed path in D.

Proof: See the argument posted in the Blackboard notes.

Definition: We say that a region D is <u>open</u> if for every point P in D there is a disk with center P that lies entirely in D. Intuitively, D is open if it does not contain any of its boundary points.

Definition: We say that a region D is <u>connected</u> if any two points in D can be joined by a path that lies in D.

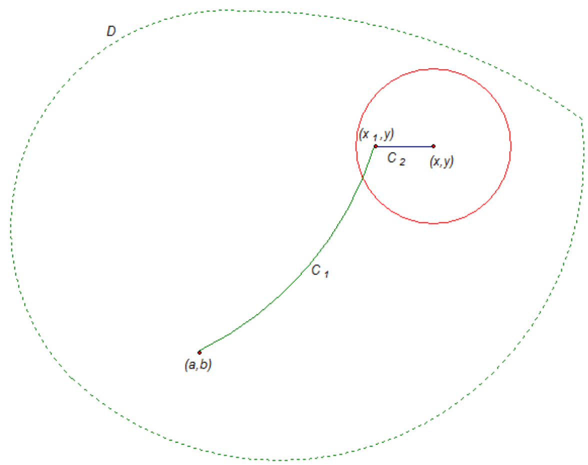
Examples:

Open	**Not Open**	**Connected**	**Not Connected**
			

Theorem (5.3.3): Suppose **F** is a vector field that is continuous on an open connected region *D*. If $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of path in *D*, then **F** is a conservative vector field on *D*.

Proof: We will prove this in the case that **F** has two components (the case where **F** has three components is similar). We need to construct a potential function for **F**. Suppose (a,b) is a fixed point in *D*. For any point (x,y) in *D* we let $f(x,y)=\int_{(a,b)}^{(x,y)} \mathbf{F}\cdot d\mathbf{r}$.

The precise meaning of the integral $\int_{(a,b)}^{(x,y)} \mathbf{F}\cdot d\mathbf{r}$ is that it represents the line integral of **F** along any path *C* which is contained in *D* and which starts at (a,b) and ends at (x,y).



Theorem (5.3.4): If $\mathbf{F}(x,y)=\langle P(x,y),Q(x,y)\rangle$ is a conservative vector field, where P and Q have continuous first order partial derivatives on a domain D, then throughout D we have $\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}$.

Question: Why does this theorem hold? (Hint: Think about the theorem from chapter 3 which told us when $f_{xy}(a,b)=f_{yx}(a,b)$)

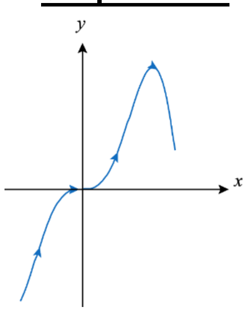
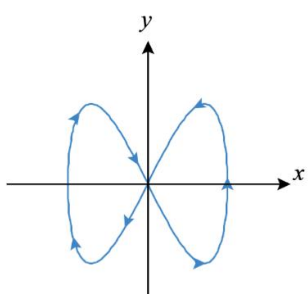
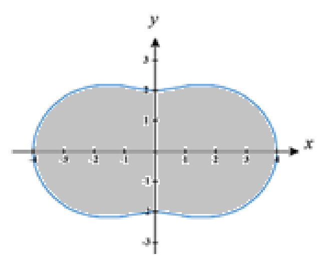
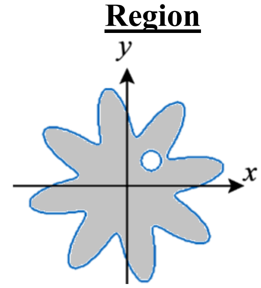
Note: From here the crucial question is: Does the converse of the above theorem hold? If it does, we would have a really nice way to check if a two dimensional vector field is conservative.

Definition: A <u>simple curve</u> is a curve which doesn't intersect itself anywhere between its endpoints.

Definition: A <u>simply-connected region</u> in the plane is a connected region D such that every simple closed curve in D encloses only points in D.

Note: Intuitively, a simply-connected region is a connected region which contains no holes.

Examples:

<u>Simple Curve</u>	**<u>Non-Simple Curve</u>**	**<u>Simply Connected Region</u>**	**<u>Not Simply Connected Region</u>**
			

Theorem (5.3.5): Let $\mathbf{F}(x,y)=\langle P(x,y),Q(x,y)\rangle$ be a vector field on an open simply-connected region D. Suppose that P and Q have continuous first order partial derivatives and $\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}$. Then, $\mathbf{F}$ is conservative.

Note: Since 2-space is an open simply-connected region, the above two theorems provide us with a very useful way for checking whether or not a vector field is conservative.

Note: An easy way to check if a vector field with three components is conservative that is similar to the above results will be shown later.

Example 6: Determine whether or not $\mathbf{F}(x,y)=\langle x-y, x-2\rangle$ conservative.

Example 7:

a) Show that $\mathbf{F}(x,y)=(3x^2-2y^2)\mathbf{i}+(-4xy+3)\mathbf{j}$ is conservative.

b) Find a potential function for **F**.

c) Suppose that the curve C is given by $\mathbf{r}(t) = \langle e^t \sin(t), e^t \cos(t) \rangle$ where $0 \le t \le \pi$. Find the work done by **F** in moving a particle along C.

Note: The method of presented in the solution to part (b) of the above example can be generalized to vector fields with three components when we know that we have a conservative vector field. In the next example assume that the vector field is conservative and try to generalize the methods of part (b).

<u>Example 8 (if time permits):</u> Find a potential function for $\mathbf{F}(x, y, z) = \langle yz + z, xz + 1, xy + x \rangle$.